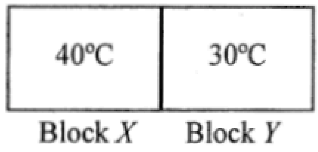
Q1

1. Two metal blocks X and Y of the same mass and of initial temperatures 40°C and 30°C respectively are in good thermal contact as shown. The specific heat capacity of X is greater than that of Y. Which statement is correct when a steady state is reached ? Assume no heat loss to the surroundings.



A. The temperature of block X is higher than that of block Y.
B. Their temperature becomes the same and is lower than 35°C.
C. Their temperature becomes the same and is higher than 35°C.
D. Their temperature becomes the same and is equal to 35°C.

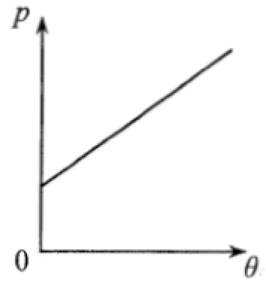
Q2

2. When a patient's arm is wiped by a piece of cotton soaked with alcohol, the wiped area will feel cool as that patch of alcohol on the skin evaporates. Which statement explains this phenomenon ?

A. The evaporation of alcohol absorbs heat from the patient's arm.
B. The alcohol on the skin releases latent heat to the surrounding air.
C. The motion of all the molecules in the patch of alcohol slows down.
D. Air molecules remove heat from the patch of alcohol by conduction.

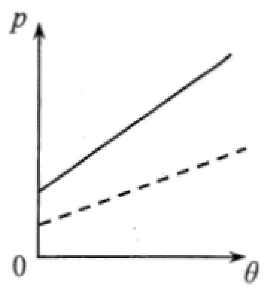
Q3

*3. An ideal gas is contained in a closed vessel of fixed volume. The graph below shows the variation of pressure p of the gas against its Celsius temperature θ.

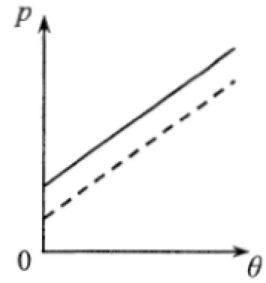


If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ ?

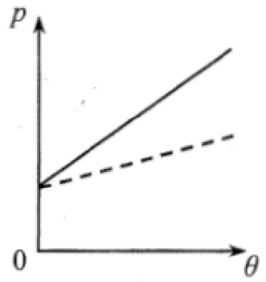
A.



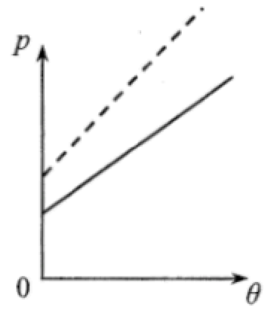
B.



C.



D.



Q4

4. Which of the following descriptions is correct ?

A. When water at 25°C is heated to 50°C, both the kinetic energy and potential energy of the water molecules increase.

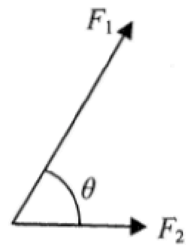
B. When water at 25°C is heated to 50°C, only the potential energy of the water molecules increases.

C. When water boils at 100°C and turns into steam, the kinetic energy of the water molecules increases.

D. When water boils at 100°C and turns into steam, the potential energy of the water molecules increases.

Q5

5.

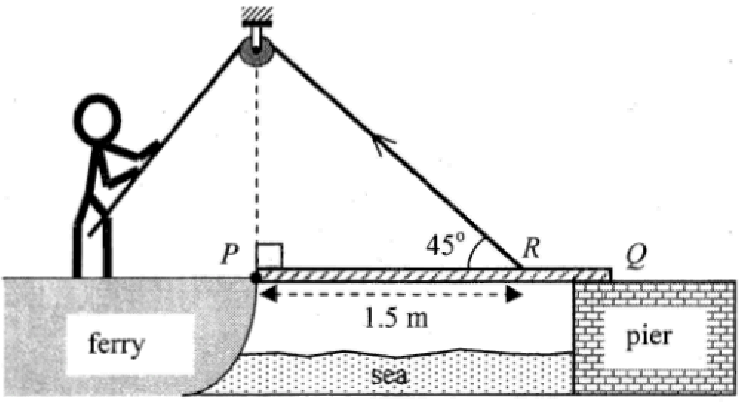


Two forces F_1 and F_2 of constant magnitudes act at the same point as shown. When the angle θ between F_1 and F_2 increases from 0° to 180°, the magnitude of the resultant force

A. decreases throughout.
B. increases throughout.
C. decreases and then increases.
D. increases and then decreases.

Q6

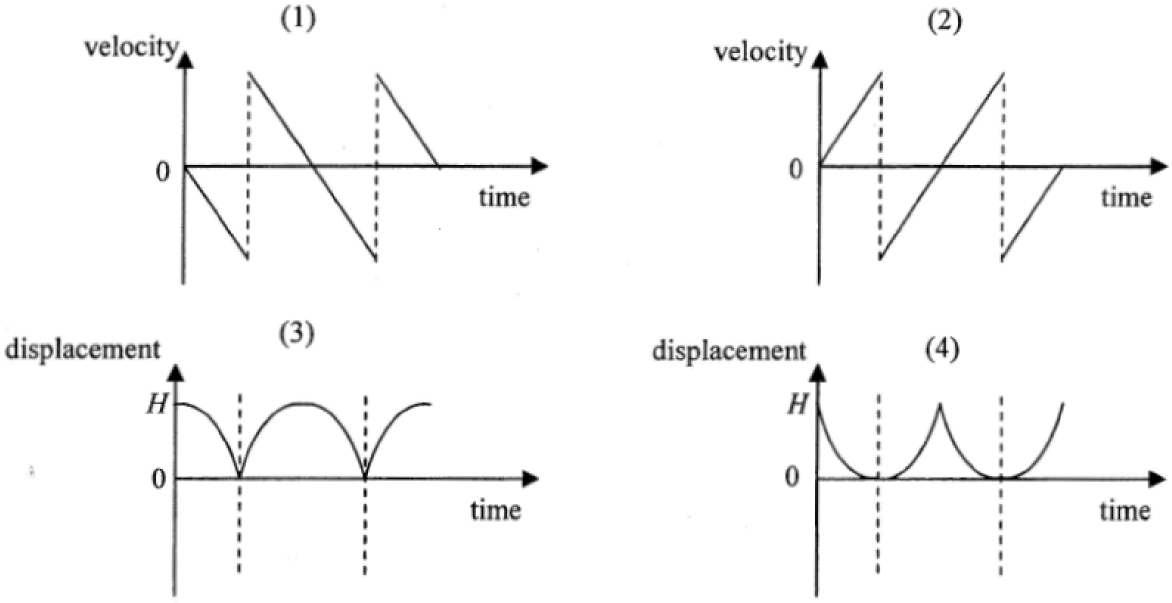
6. A uniform gangplank PQ of a ferry smoothly hinged at end P initially rests horizontally on the pier. The gangplank has mass M and length 2 m. It is raised by a man on the ferry using a light rope passing a smooth fixed light pulley and connecting to R on the gangplank as shown. R is 1.5 m from end P. Which of the following correctly describes the force required to raise the gangplank steadily ?



	initial force required to raise the gangplank when it is horizontal	*subsequent* force required to raise the gangplank
A.	0.67 Mg	greater than 0.67 Mg
B.	0.67 Mg	smaller than 0.67 Mg
C.	0.94 Mg	greater than 0.94 Mg
D.	0.94 Mg	smaller than 0.94 Mg

Q7

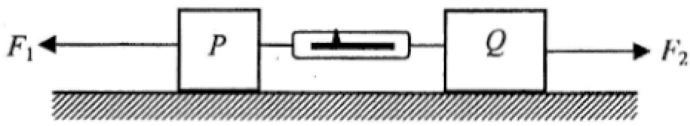
7. Which of the following graphs (velocity-time and displacement-time) best represent the motion of a ball falling from rest under gravity at a height H and bouncing back from the ground two times ? Assume that the collision with the ground is perfectly elastic and neglect air resistance. (Downward measurement is taken to be negative.)



A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

Q8

8.



Blocks P and Q of mass m and $2m$ respectively are connected by a light spring balance and placed on a smooth horizontal surface as shown. If horizontal forces F_1 and F_2 (with $F_1 > F_2$) act on P and Q respectively and the whole system moves to the left with constant acceleration, what is the reading of the spring balance ?

A. $\dfrac{2F_1 - F_2}{3}$

B. $\dfrac{2(F_1 - F_2)}{3}$

C. $\dfrac{2F_1 + F_2}{3}$

D. $\dfrac{F_1 + 2F_2}{3}$

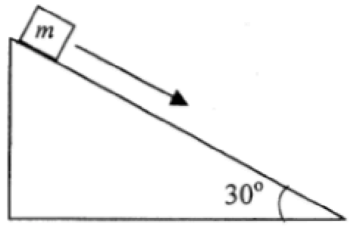
Q9

9. An object of mass 0.5 kg is raised vertically from the ground by a motor. The object is raised 2.5 m in 1.5 s with uniform speed. Estimate the output power of the motor. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

A. 5.5 W
B. 8.2 W
C. 11.0 W
D. 16.4 W

Q10

10. A block of mass m resting on a 30° incline is given a slight push and slides down the incline with a uniform speed. Which of the following statements about the block's motion on the incline is/are correct ?

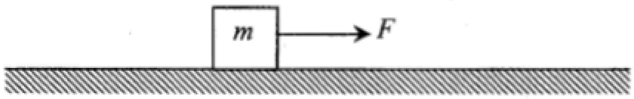


(1) There is no net force acting on the block.
(2) The frictional force acting on the block is 0.5 mg.
(3) If the block is given a greater initial speed, it will slide down the incline with acceleration.

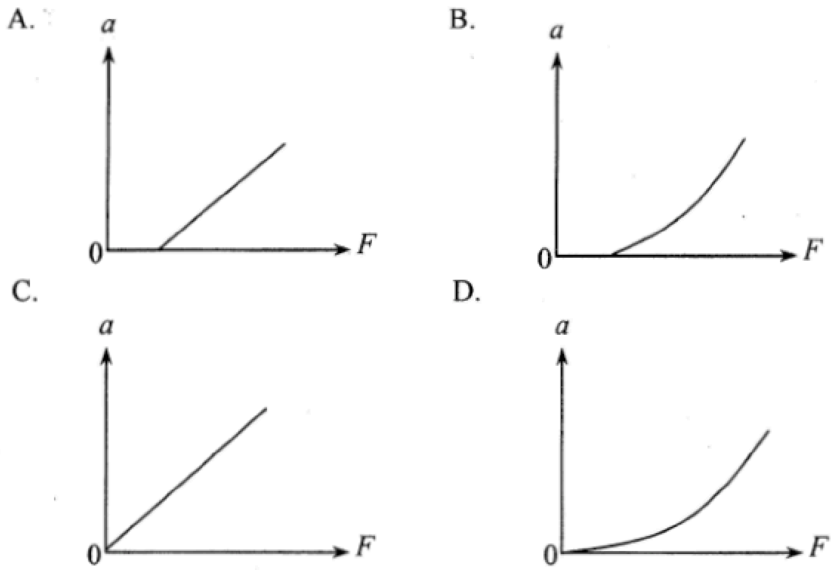
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q11

11.



A block of mass m initially resting on a rough horizontal surface is pulled along the surface by a horizontal force F increasing from zero. If the frictional force is constant, which graph shows the relation between the acceleration of the block a and force F ?



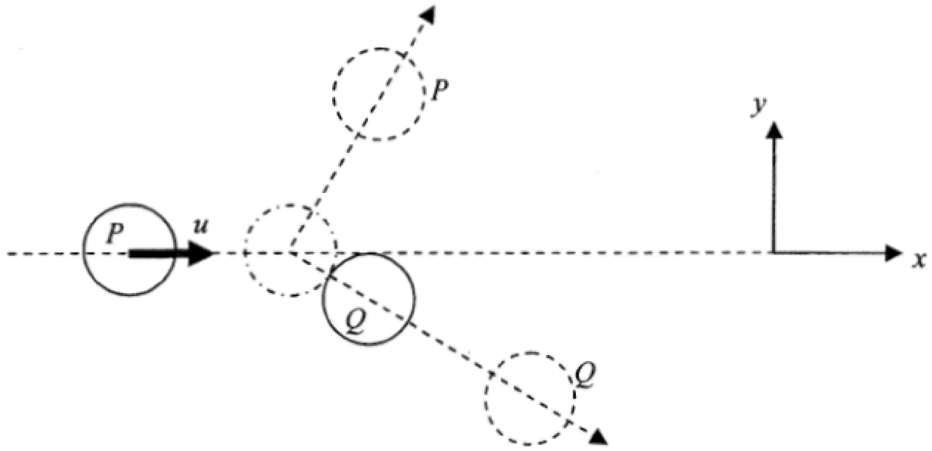
Q12

*12. A bomber aircraft is 1 km above the ground and is flying horizontally at a speed of 200 m s^{-1}. The aircraft is going to release a bomb to destroy a target on the ground. How long before flying over the target should the bomb be released ? Assume that the bomber aircraft and the target are in the same vertical plane and neglect air resistance. ($g = 9.81$ m s^{-2})

A. 5.6 s
B. 10.1 s
C. 14.3 s
D. It cannot be calculated as the horizontal distance between the aircraft and the target is not known.

Q13

13. On a smooth horizontal surface, a circular disk P moving at velocity u along the x direction collides obliquely with an identical disk Q initially at rest as shown below. The mass of each disk is m. Which statements about the collision is/are correct ?



(1) Momentum of the system in the y direction is not conserved.

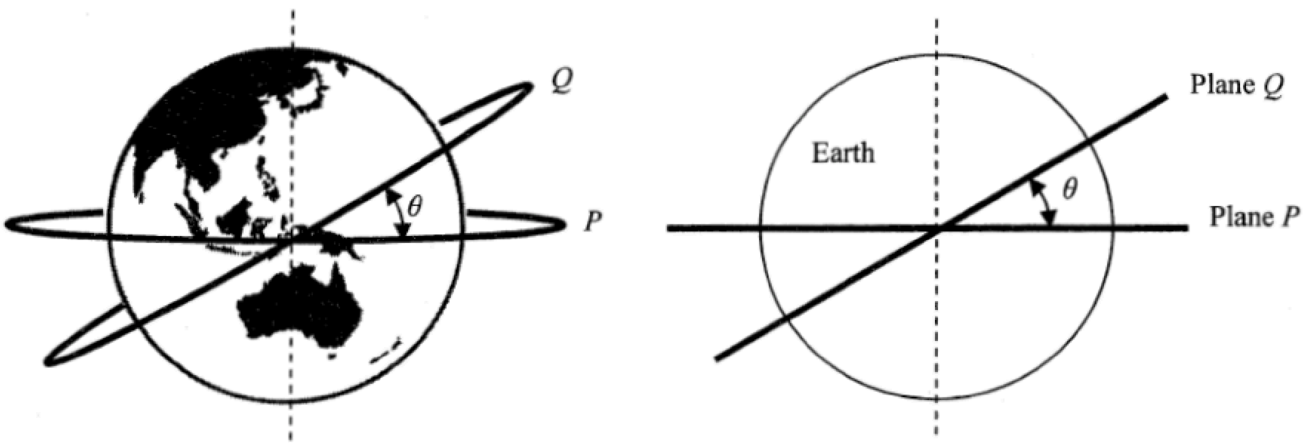
(2) The total kinetic energy of P and Q after collision is $\frac{1}{2}mu^2$ if the collision is perfectly elastic.

(3) Speed of Q after collision is less than u.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q14

*14. Two satellites move in circular orbits of the same radius R around the Earth (mass M). The orbits are in two different planes P and Q as shown. Plane P coincides with the Earth's equator while plane Q is inclined to the equator at θ. Which statement is **INCORRECT** ?



A. The speed of satellite P is $\sqrt{\frac{GM}{R}}$.

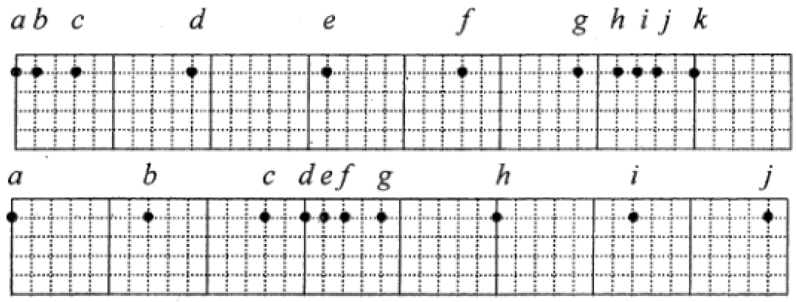
B. The centripetal force acting on satellite Q is pointing along the plane Q.

C. The acceleration of both satellites is the same in magnitude.

D. The period of satellite Q is longer than that of satellite P.

Q15

15.

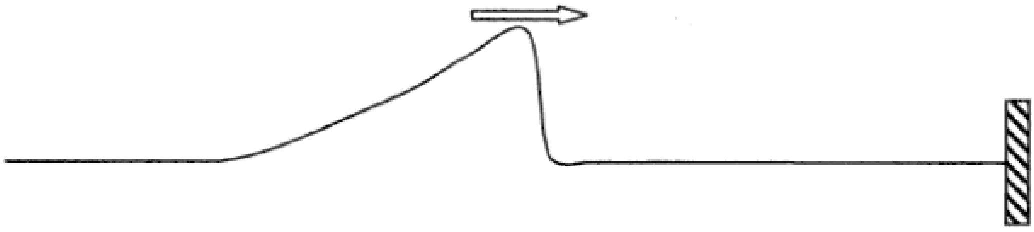


A series of particles is uniformly distributed along a slinky spring initially. Figure (a) shows their positions at a certain instant when a travelling wave propagates along the slinky spring from left to right. Figure (b) shows their positions 0.1 s later. Which statement is correct ?

A. Particle *e* is always stationary.
B. Particles *a* and *i* are in phase.
C. The wavelength of the wave is 16 cm.
D. The frequency of the wave is 10 Hz.

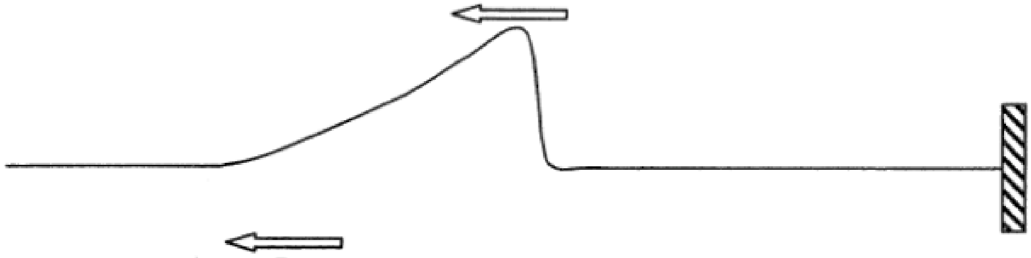
Q16

16. A pulse on a string propagates towards the right end which is fixed.

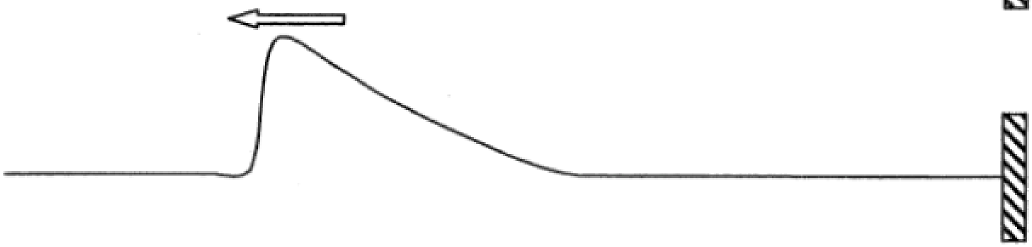


Which of the following represents the reflected pulse ?

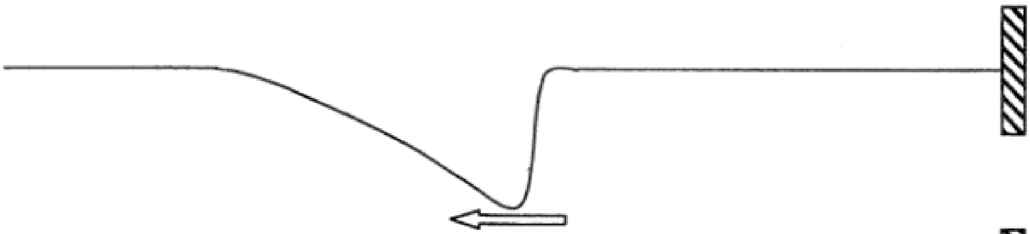
A.



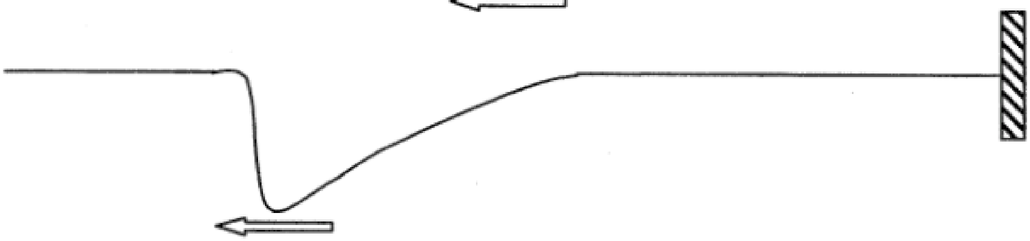
B.



C.

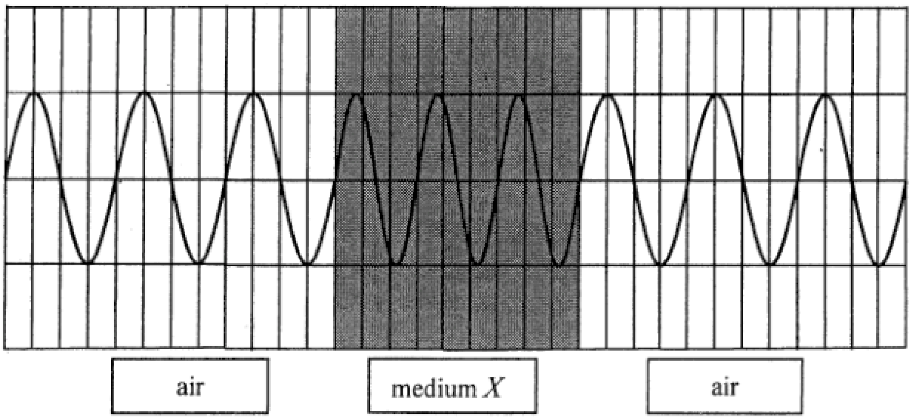


D.



Q17

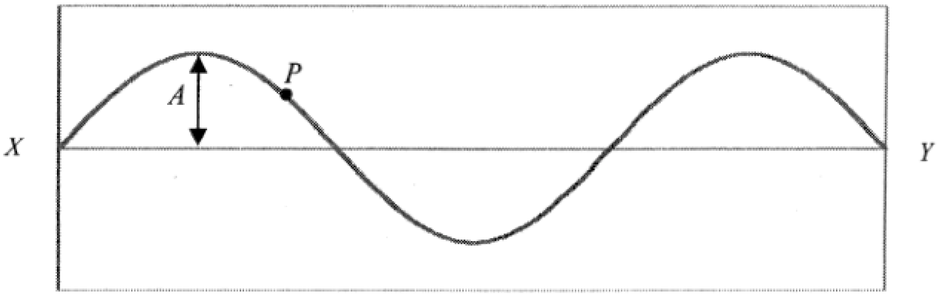
17. A certain monochromatic light passes through medium X as shown below. What is the refractive index of medium X ?



A. 1.25
B. 1.33
C. 1.50
D. 1.65

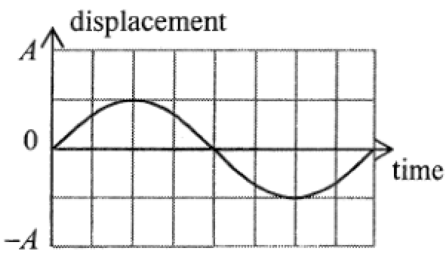
Q18

18. A stationary wave is formed on a string fixed at both ends X and Y. The following is a snapshot of the string at time $t = 0$. The amplitude of vibration at an antinode is A.

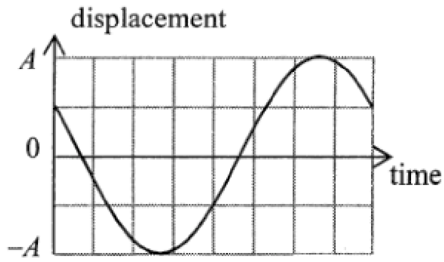


Which of the following shows the displacement-time graph of point P on the string for one period ? (Upward displacement is taken as positive.)

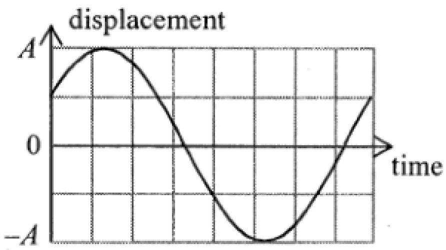
A.



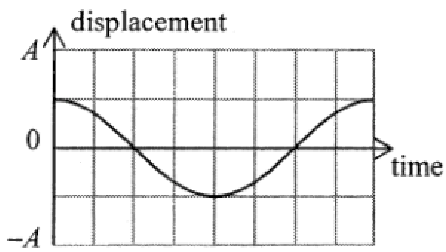
B.



C.



D.



Q19

19. Which of the following statements is **INCORRECT** ?

A. In air, the wavelength of infra-red radiation is shorter than that of ultra-violet radiation.
B. Visible light travels faster in air than in glass.
C. Microwaves travel at the speed of light in a vacuum.
D. Both light and sound exhibit diffraction.

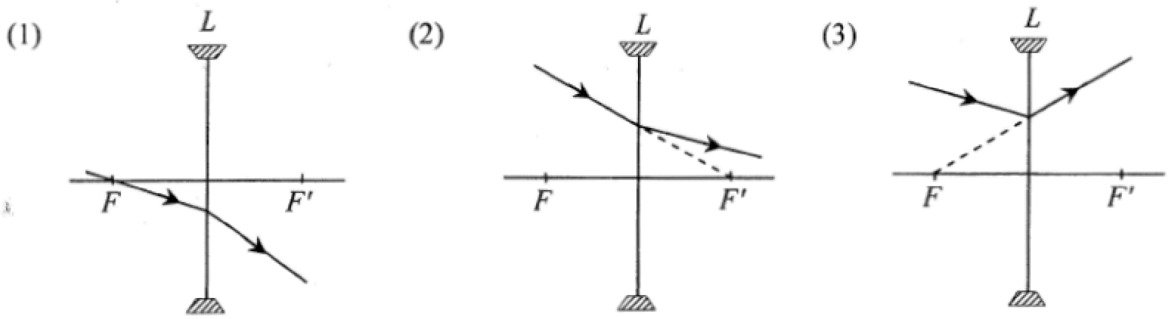
Q20

*20. For a diffraction grating of 600 lines per mm, the diffracted red light (657 nm) coincides with the diffracted violet light (438 nm) at angle of diffraction 52°. What are the respective orders of the diffracted red light and violet light ?

	red	**violet**
A.	2	3
B.	3	4
C.	3	2
D.	4	3

Q21

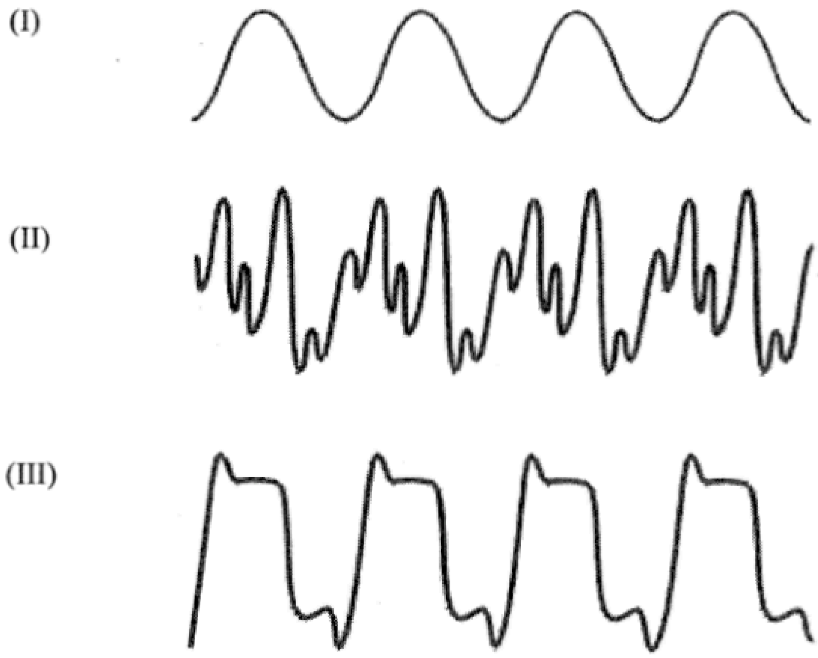
21. In each of the following diagrams, L is a concave lens and its two principal foci are denoted by F and F'. Which of the ray diagrams is/are possible ?



A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q22

22. The figure shows the waveforms of sound notes generated by a violin, a piano and a tuning fork. The scale is the same in time and intensity axes for all three waveforms.



Which of the following about the sound notes are correct ?

(1) They all have the same pitch.
(2) The qualities of sound of (II) and (III) are different.
(3) (I) is generated by the tuning fork.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

Q23

23. Which of the following about ultrasound is **INCORRECT** ?

A. Ultrasound is a longitudinal wave.
B. The frequency of ultrasound is greater than 20000 Hz.
C. In air, the speed of ultrasound is faster than the speed of audible sound.
D. In air, the diffraction effect of ultrasound is less prominent than that of audible sound.

Q24

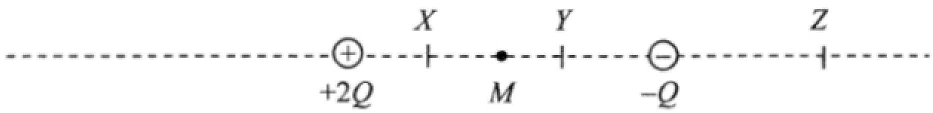
24. P, Q, R, S are charged objects. When two of them are brought close to each other, P and Q repel, R and S also repel while Q and R attract each other. Which of the following descriptions about their charges is/are possible ?

(1) P and R are negatively charged.
(2) Q and S are positively charged.
(3) P is positively charged and S is negatively charged.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q25

*25.



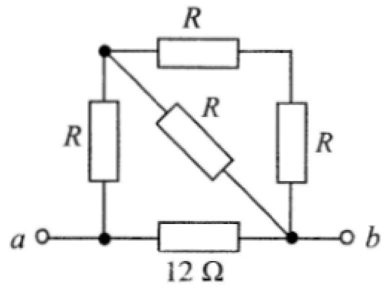
Two point charges $+2Q$ and $-Q$ are situated at fixed positions as shown. M is the mid-point between the charges. X, Y and Z are points marked on the line joining these two charges. At which point could

(1) the resultant electric field due to the two charges be zero ?
(2) the total electric potential due to the two charges be zero ?

	(1)	(2)
A.	Z	X
B.	Z	Y
C.	X	Z
D.	Y	Z

Q26

26.

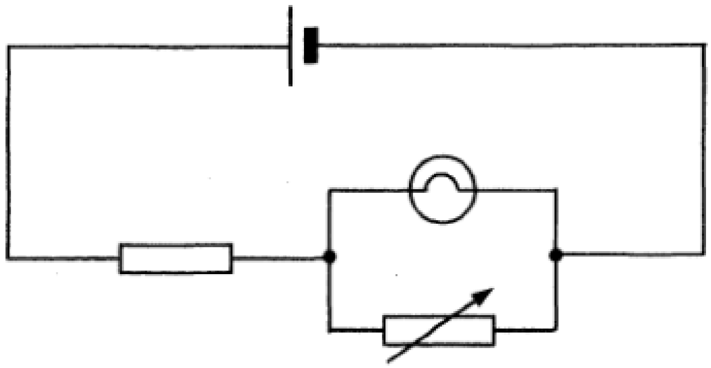


In the above network, the resistance across terminals a and b is 6 Ω. If the 12 Ω resistor is replaced by a 6 Ω resistor, the resistance across terminals a and b

A. becomes 2 Ω.
B. becomes 4 Ω.
C. becomes 6 Ω.
D. cannot be found as the value of R is unknown.

Q27

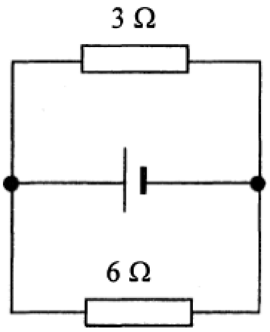
27. What will happen if the variable resistor is set to zero in the circuit below ?



A. The light bulb will burn out.
B. The light bulb will not light up.
C. The brightness of the light bulb will increase.
D. The brightness of the light bulb will remain unchanged.

Q28

28.

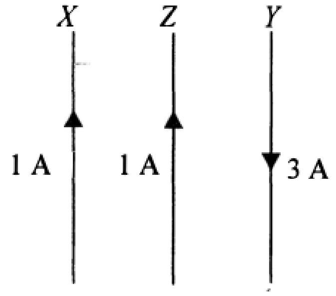


In the above circuit, the cell has e.m.f. 12 V and internal resistance 2 Ω. What is the current in the 6 Ω resistor ?

A. 0.5 A
B. 1.0 A
C. 1.5 A
D. 2.0 A

Q29

29. In the figure below, X, Y and Z are three long straight parallel wires with Z placed midway between X and Y. X and Z carry currents of 1 A in the same direction while Y carries a current of 3 A in the opposite direction. The magnetic force per unit length experienced by wire X due to wire Z is of magnitude F.



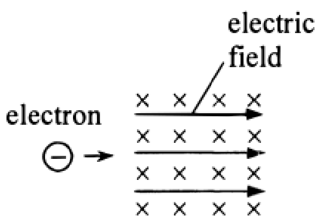
The magnetic force per unit length acting on wire Z due to both X and Y is

A. $2F$ to the right.
B. $2F$ to the left.
C. $4F$ to the right.
D. $4F$ to the left.

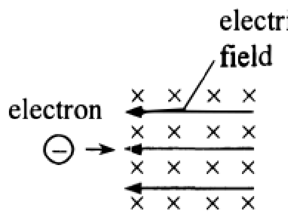
Q30

30. An electron enters a region in which both a uniform electric field E and a uniform magnetic field B exist. The magnetic field B is pointing into the paper. In which direction should the electric field be applied so that the electron could be undeflected ?

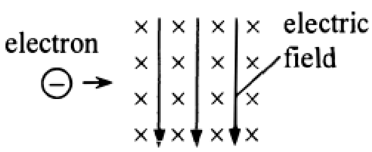
A.



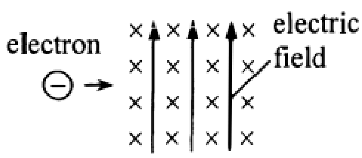
B.



C.

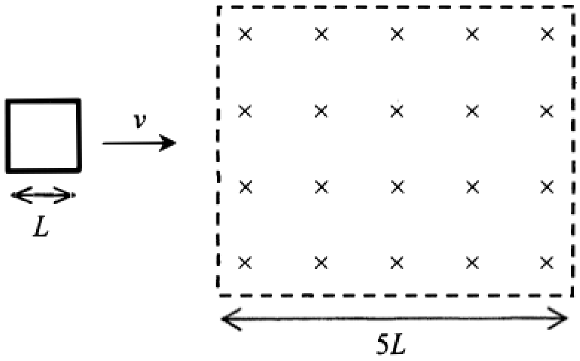


D.



Q31

31.



A square metal frame of length of side L moving with constant velocity v passes through a region of uniform magnetic field of width $5L$ as shown. What is the total time period during which a current is induced in the frame ?

A. $\frac{L}{v}$

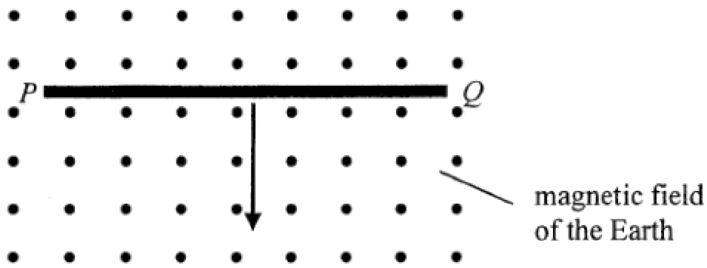
B. $\frac{2L}{v}$

C. $\frac{3L}{v}$

D. $\frac{4L}{v}$

Q32

32. A copper rod PQ is placed horizontally as shown below. It is released and then falls vertically, cutting across the magnetic field of the Earth pointing out of the paper. Neglect air resistance. Which of the following statements is/are correct ?



(1) A voltage is induced across PQ.

(2) A steady induced current is generated in the rod.

(3) Due to the effect of the Earth's magnetic field, the copper rod falls with an acceleration less than the acceleration due to gravity.

A. (1) only

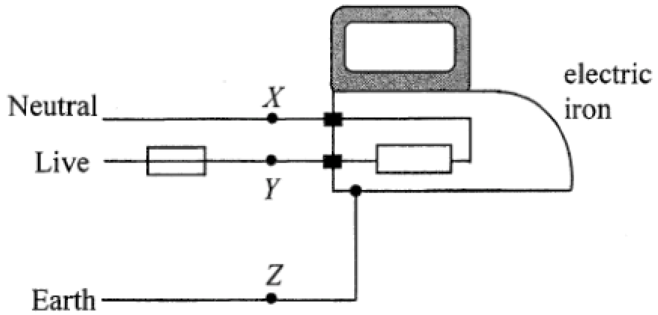
B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Q33

33.



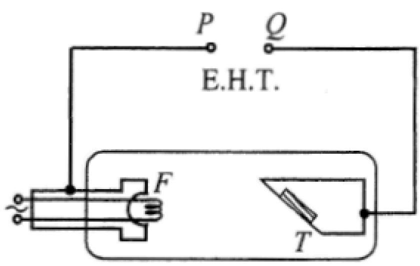
The figure shows a simple domestic circuit for an electric iron. The fuse will blow when which of the following points are short-circuited ?

(1) X and Y
(2) Y and Z
(3) X and Z

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

Q34

34.



The figure shows a schematic diagram of an X-ray tube in which the filament F and the metal target T are connected to terminals P and Q of an E.H.T. Which statement is correct ?

A. P is the positive terminal and X-rays are emitted from T.
B. P is the positive terminal and X-rays are emitted from F.
C. Q is the positive terminal and X-rays are emitted from T.
D. Q is the positive terminal and X-rays are emitted from F.

Q35

35. A certain radioactive isotope X has a half-life of 20 hours. After a time interval of 10 hours, what is the approximate fraction (f) of a sample of the radioactive isotope X remaining ?

A. $\frac{1}{4} \le f \le \frac{1}{2}$

B. $f = \frac{1}{2}$

C. $\frac{3}{4} > f > \frac{1}{2}$

D. $f > \frac{3}{4}$

Q36

36. Isotopes of an element have different mass number A and neutron number N. Which of the following N - A plots correctly shows the relationship of N and A for any given element ?